

Simon Sobti

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Professional Summary

Computational bioinformatics scientist (PhD, UCSF) with 9+ years of experience in multi-omics data integration, machine learning, image analysis, and pipeline development for cancer genomics (and 5+ years of wet lab experience). Expertise spans single-cell RNA-seq, scATAC-seq, spatial transcriptomics, and bulk RNA-seq analysis, with a strong record of translating high-dimensional genomic data into actionable biological insights and therapeutic targets. Proficient in Python, R, PyTorch, and cloud-based HPC environments (AWS). Co-author on 4 peer-reviewed and preprint publications with two first author publications; experienced in cross-functional collaboration with wet-lab and clinical teams.

Education

University of California, San Francisco (UCSF) 2018-2025

PhD in Biochemistry and Molecular Biology (April 2025)

Thesis: Computational methods to decipher targetable transcriptional states in breast cancer

University of Missouri 2013-2016

B.S., Chemistry (May 2016)

Minors in Biology and Mathematics

Skills

- scRNA-seq (Seurat, scanpy, etc.)
- scATAC-seq (ArchR, Signac, SnapATAC)
- Spatial Transcriptomics (10x Visium, CosMx)
- Nextflow pipelines for end-to-end analysis (Docker + AWS EC2, Batch, S3)
- Cell culture, molecular cloning, Illumina NGS library prep (5+ years wet lab experience)
- Patient bulk RNA-seq (clinical data)
- Whole genome sequencing (WGS)
- Image analysis (CellProfiler + ImageJ)
- Statistics (generalized linear model regression, PCA, curve fitting, etc.)
- Cancer target discovery
- R/Python and ggplot2
- Unix/Bash + HPC
- CRISPR editing
- Deep learning (PyTorch, Lightning, Hugging Face) + Variational Autoencoders (VAEs)
- Fine-tuned Geneformer (state of the art models for single cell data) for perturbation inference
- Scikit-learn
- Science communication: 6+ poster presentations at national conferences (Cold Spring Harbor, NIH, NCI); experienced presenting computational findings to interdisciplinary audiences

Research Experience

Cancer bioinformatics

09/2018-04/2025

Graduate Student at UCSF in Hani Goodarzi's lab

- Developed and open-sourced VariPath, an R package for pathway activity scoring in Perturb-seq data, outperforming comparable methods in signal-to-noise ratio for perturbation-pathway association (manuscript in preprint).
- Developed and open-sourced txnheterogeneity, an R package for quantifying transcriptome-wide expression heterogeneity and open chromatin heterogeneity between experimental conditions in patients and cells.
- Validated role of MIS18A and RNF8 as drivers of transcriptional heterogeneity in scATAC-seq data by identifying changes in global heterogeneity of open chromatin.
- Analyzed bulk RNA-seq data from hundreds of TCGA breast cancer patients to uncover novel drivers of intratumoral transcriptomic heterogeneity linked to poor clinical outcomes, integrating patient clinical metadata with gene expression profiles for survival association analysis.
- Designed and trained a PyTorch variational autoencoder (VAE) for dimensionality reduction of scRNA-seq data across a miRNA/tRNA-fragment perturbation library in breast cancer cells, enabling systematic identification of transcriptional state changes associated with small ncRNA activity.
- Quantified cell shape differences using CellProfiler to measure transcriptional heterogeneity's effect on cell and organelle morphology features from CellPAINT data. Experience with ImageJ as well as prepping and imaging FISH samples with confocal microscopy from other projects. Familiarity with MERFISH from work of others in the Goodarzi lab.
- Led project to perform spatial transcriptomic analysis (in collaboration with Farnaz Shamsi's lab) of 10x Visium slides of mouse and vervet monkey adipose tissue to identify progenitor cell states surrounding white adipose tissue that lead to shifts in brown adipose tissue content under varying conditions.
- Benchmarked and deployed scikit-learn classifiers (regression trees, SVM, random forest) for cross-modality transfer of drug response labels (GI50) from bulk RNA-seq cell line data to individual cells in scRNA-seq, enabling single-cell drug sensitivity profiling — a key approach for precision oncology target identification.
- Fine-tuned Geneformer (a single-cell foundation model) via Hugging Face Transformers for perturbation inference tasks on in-house datasets; built LangChain-based retrieval-augmented generation (RAG) pipelines using LLM APIs to enable natural language querying of collected experimental data.
- Mapped single cell resolution transcriptomic effects from scRNA-seq data and identified targetable vulnerabilities, specifically drug/gene target combinations in triple negative breast cancer cell lines. Identified RNF8 and MIS18A and separately the cGAS/STING pathway as targets for improved combinatorial chemotherapy response to 5FU and cyclophosphamide or gemcitabine and doxorubicin, respectively.
- Analyzed Perturb-seq data from miRNA and tRNA fragment inhibition library in MDA-MB-231 cells and identified novel roles that many tRNA fragments play that previously were thought to only be biomarkers in breast cancer.

RNA Biology

09/2016-07/2018

Postbaccalaureate Researcher in Astrid Haase's lab at the NIH campus in Bethesda, MD

- Generated and analyzed PAR-CLIP small RNA libraries from *Drosophila* protein tag pull downs.
- Edited ovarian somatic stem cells from *Drosophila Melanogaster* using CRISPR and drug selection.

Publications and Poster Presentations

- Choi B*, Sobti S*, Soto LM, Charbonneau T, Sababi A, Navickas A, Najafabadi HS, Goodarzi H. Decoy-seq unlocks scalable genetic screening for regulatory small noncoding RNAs. bioRxiv. 2025.01.25.634869; doi: <https://doi.org/10.1101/2025.01.25.634869>.
- Woo BJ*, Sobti S*, Yousefi H, Garcia K, Zhou S, Borah A, Goodarzi H. Systematic identification of chromatin organizers as stochastic tuners of intratumoral transcriptional heterogeneity. bioRxiv. 2026.04.18.719392; doi: <https://doi.org/10.64898/2026.04.18.719392>.
- Jacobo Jacobo M*, Donnella HJ*, Sobti S, Kaushik S, Goga A, Bandyopadhyay S. An inflamed tumor cell subpopulation promotes chemotherapy resistance in triple negative breast cancer. Sci Rep. 2024 Feb 14;14(1):3694. doi: 10.1038/s41598-024-53999-w.
- Stein CB*, Genzor P*, Mitra S*, Elchert AR*, Ipsaro JJ, Benner L, Sobti S, Su Y, Hammel M, Joshua-Tor L, Haase AD. Decoding the 5' nucleotide bias of PIWI-interacting RNAs. *Nat Commun* 10, 828 (2019). <https://doi.org/10.1038/s41467-019-08803-z>
- Sobti S, Sarshad AA, Genzor P, Guydosh NR, Hafner M, Haase AD. Characterizing constituents and patterns of the piRNA biogenesis machinery. Poster presented at: Regulatory & Non-Coding RNAs; 2018 May 15-19; Cold Spring Harbor, NY.
- Sobti S, Genzor P, Stein CB, Mitra S, Haase AD. Characterization of PIWI-interacting small RNA biogenesis. Poster presented at: Postbac Poster Day 2017; 2017 May 4; Bethesda, MD.
- Stein CB, Sobti S, Mitra S, Ipsaro JJ, Joshua-Tor L, Haase AD. Dissecting the 5'U bias of piRNAs. Poster presented at: National Cancer Institute Symposium RNA Biology 2017; 2017 Apr 24-25; Bethesda, MD.
- Sobti S, Genzor P, Stein CB, Mitra S, Haase AD. Characterization of PIWI-interacting small RNA biogenesis. Poster presented at: 12th Annual NIDDK Scientific Conference; 2017 Apr 13-14; Bethesda, MD.

(*: denotes co-authorship)